

Who Can Be Read? An Accountability Framework and Claims Audit for Individual Neuro-Inference

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Abstract

Products and clinical tools increasingly claim to infer an individual's emotion, attention, or likely behaviour from brain and body signals, and increasingly from a *predicted* brain, where a model maps a stimulus to cortical activity that is then read as a measurement. These claims are sold to advertisers, employers, and clinicians, and they assert private knowledge about specific people, yet there is no shared standard for deciding which of them can be true. We supply one. We distinguish three claim types, population, identity, and individual content, and show they have sharply different evidentiary status: a simple and provable boundary, the individuation gap, makes individual-content claims from a zero-shot predicted brain unsupportable, while identity and population claims are supportable under stated tests. From this we build three accountability artifacts. First, an audit rubric that classifies a marketed claim and assigns it a test and a verdict. Second, a mapping of the three claim types onto existing regulation, including the European Union's rules on emotion recognition and biometric categorization, data-protection law, and consumer-protection enforcement, which shows that the claims least supportable by evidence are also the ones regulation most restricts. Third, a disclosure standard and a refusal-first deployment pattern that keep identity recognition, which works, separate from mental-content reading, which does not. The contribution is a concrete way to hold individual neuro-inference claims accountable, grounded in a verifiable technical boundary rather than in contested notions of harm.

Keywords

neurotechnology, individual inference, accountability, transparency, auditing, emotion recognition, biometric categorization, information asymmetry, EU AI Act, disclosure

1 Introduction

Systems that infer facts about people are the core concern of fairness, accountability, and transparency, and a fast-growing class of them infers mental content. Affective-computing products score emotion from a face or a voice, brand-intelligence platforms sell predicted audience response, and a new wave of neuro-mental-health tools offer to read an individual's state from brain or body signals. Many do so from a *predicted* brain: a model maps a stimulus or context to cortical activity, which is then labelled with construct names and reported as if it had been measured [4, 8, 9].

The accountability problem is that these individual claims are asserted, priced, and acted on before anyone has agreed how to tell which are true. A person scored by such a system cannot easily contest the score, and a buyer or regulator cannot easily separate a claim that the evidence supports from one that it does not, because both arrive dressed as neural measurements. The result is an information asymmetry: the operator holds what looks like privileged

knowledge about a specific person, and the person and the public hold no means to check it.

This paper gives a means to check it. We separate three kinds of claim and show they are not interchangeable. A *population* claim predicts an aggregate or an audience. An *identity* claim recognizes or verifies which person is present. An *individual-content* claim states what a specific person feels, attends to, or will choose. A short and verifiable technical boundary, which we call the individuation gap, makes the third kind unsupportable for a predicted brain that has never seen the person, while the first two are supportable under tests we state. On this boundary we build an audit rubric, a mapping to existing law, and a disclosure standard. Our aim is not to adjudicate whether neurotechnology is good or bad, but to give regulators, auditors, and builders a concrete and falsifiable basis for deciding which individual claims may be made.

Contributions. (1) A taxonomy of individual neuro-inference claims with, for each type, the test that would support it and its current evidentiary status. (2) An audit rubric that takes a marketed claim to a verdict (Figure 1). (3) A mapping of the taxonomy onto emotion-recognition, biometric-categorization, data-protection, and consumer-protection regimes (Table 1). (4) A disclosure standard and a refusal-first deployment pattern that separate identity from content.

2 Background: the individuation gap

We state the boundary compactly; it is elementary and is proved, with machine-checked support, in a companion technical analysis. Let an outcome be $Y = S + \epsilon$ with idiosyncratic-noise variance $v > 0$, and let a predictor resolve an identifiable *between-person* signal variance $a \geq 0$. The best-case squared correlation is $\rho^2 = a/(a + v)$. Aggregating over N people replaces v with v/N and so raises ρ^2 ; this is the population regime. A zero-shot predicted brain, by contrast, emits the same features for every unseen person, so it carries no between-person variance, $a = 0$, and $\rho^2 = 0$. The individual signal is not merely hard to extract; it is absent from the input, so no model and no amount of pooling recovers it. Per-person enrollment can supply $a > 0$, but on current encoders and signals that enrollment recovers a person's *identity* far more readily than their *behaviour*: in-scanner fingerprinting reaches 100% and a behavioural fingerprint reaches many times chance, while the direct link from an individual's neural signal to that individual's behaviour abstains under a calibrated test. The measurement literature agrees: individual brain-behaviour effects are small and need very large samples, individual test-retest reliability is low, and individual precision mapping takes hours of data per person [2, 3, 5, 7]. These facts are the premise of this paper, not its contribution; what follows builds accountability on top of them.

Auditing an individual neuro-inference claim

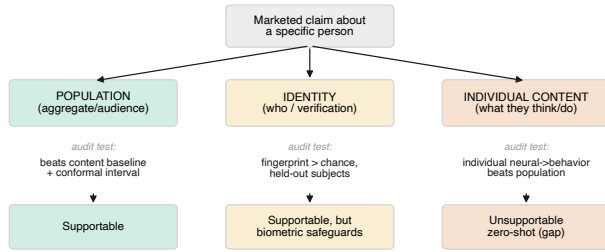


Figure 1: The audit rubric. A marketed claim about a specific person is classified as population, identity, or individual-content; each type has an explicit test and a verdict. Individual-content claims from a zero-shot predicted brain fail by the individuation gap; identity claims pass but trigger biometric safeguards; population claims pass under a content-baseline and calibration test.

3 A taxonomy of individual neuro-inference claims

We separate three claim types by what they assert about a person.

Population. The claim is about an aggregate, an audience, or a market, not about any one person. It is supportable when the predicted index beats a content-only baseline out of sample and carries a calibrated interval. This is the regime where classical neuroforecasting works [1, 6].

Identity. The claim recognizes or verifies which person is present. It is supportable when a fingerprint exceeds chance on held-out data. It is also, precisely because it works, a biometric capability that carries re-identification and surveillance risk.

Individual content. The claim states what a specific person feels, attends to, or will do. From a zero-shot predicted brain it is unsupported by the individuation gap, and even with enrollment it is at present modest and expensive, and must be shown to beat a population model for that person.

The three are routinely conflated. A system that fingerprints a person well (identity) is advertised as reading that person’s mind (content), and a system that forecasts an audience well (population) is sold as reading the individuals in it. The taxonomy makes the substitution visible and testable.

4 An audit rubric

Figure 1 turns the taxonomy into a procedure an auditor can run on a product claim without access to the vendor’s model. The auditor first classifies the claim by what it asserts about the person, then applies the corresponding test, then records the verdict and the required safeguards. The rubric is deliberately conservative: a claim that cannot be assigned a passing test defaults to *unsupported*, so the burden of evidence rests on the operator. The rubric also separates the channels, so a vendor may be credited for a working identity or population capability without being credited for an unsupported content capability.

Table 1: The claim taxonomy against existing regulation. The least evidence-supportable claim type (individual content) is also the most restricted.

Claim type	Evidentiary status	Regulatory exposure
Population	Supportable with test	Advertising/consumer law on efficacy claims
Identity	Supportable; biometric	Biometric identification rules; data-protection (biometric data)
Individual content	Unsupported zero-shot	Emotion-recognition limits; deceptive-claim enforcement

A practical consequence is that much of what is marketed as individual mind-reading is, on audit, either a population instrument (useful, but not about the individual) or an identity instrument (about the individual, but recognition rather than mind-reading). Naming which of the three a product actually delivers is most of the accountability work.

5 Mapping to regulation

The taxonomy aligns with where regulation already draws lines (Table 1). The European Union’s AI regulation restricts emotion inference in workplaces and schools and restricts biometric categorization, which maps onto our identity and individual-content rows; data-protection law treats biometric and health-adjacent inference as sensitive; and consumer-protection enforcement reaches deceptive efficacy claims, which is exactly what an unsupported individual-content claim is. The alignment is not a coincidence: the claims that the evidence least supports are the ones that most directly assert private knowledge about a person, which is what these regimes guard.

6 A disclosure standard and a refusal-first pattern

Accountability needs an artifact that survives deployment. We propose two. A *disclosure standard* requires an operator to state, for each reported score, which of the three claim types it is, which test it passed, and the interval it carries; a score that cannot name its type or test is disclosed as unsupported. A *refusal-first* deployment pattern returns an individual-content score only when the construct has a validated localizer, the predicted index beats a content-only baseline, and the score carries a conformal interval, and otherwise returns nothing. Identity and content are reported on separate, separately-governed channels, so a system may verify a person without asserting that it has read their mind. Together these move the default from assert-by-default to withhold-by-default, which is the posture the evidence warrants.

7 Limitations

The boundary we build on is a best-case abstraction; richer encoders change its constants but not its sign, since a zero-shot predictor carries no between-person variance. The regulatory mapping is a

structural alignment, not legal advice, and specific obligations vary by jurisdiction and change over time. The audit rubric classifies claims, not internal model quality, and a vendor could pass the identity test while still making unsupported content claims elsewhere, which is why the disclosure standard is needed alongside it. The empirical figures quoted as premises are drawn from existing public datasets and from the cited literature, and the companion technical analysis should be consulted for their derivation.

Generative AI Disclosure

Generative AI tools were used to assist with drafting and editing the text and with writing figure code. All research questions, framing, analysis, and claims are the author’s own, and the author takes full responsibility for the content of this paper.

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